

Routing Protocols

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Static and Dynamic Routes

■ Static routes

- The network administrator manually enters the routing information in the router.
- based upon the final packet destination.

■ Dynamic routes

- Routers can learn the information from each other on the fly.
- Using **routing protocol** to update routing information.
 - EGP, BGP, RIP, IGRP, EIGRP, OSPF ...

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Static vs. dynamic routes

■ Static routes

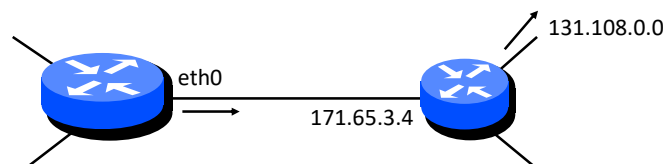
- When there is only one path to a destination network.
- For hiding parts of an intern-network.
- To test a particular link in a network.

■ Dynamic routes

- Automatic maintenance of routing table.
- Relies on routing protocol to share knowledge.
- Timely distribution of routing information and updates.
- Routers can adjust to changing network conditions.

Static routes

- `route add 192.168.1.0/24 dev eth0`
- `ip route 10.0.0.0 255.0.0.0 serial 3`
- `ip route 131.108.0.0 255.255.0.0 171.65.3.4`



Dynamic Routing Protocols

- The **purpose of a dynamic routing protocol** is to:
 - Discover remote networks
 - Maintaining up-to-date routing information
 - Choosing the best path to destination networks
 - Ability to find a new best path if the current path is no longer available
- Components of routing protocol
 - Algorithm
 - used for best path determination
 - Routing protocol messages
 - Used for discovering neighbors and exchange of routing information

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Dynamic Routing Protocols

- **Advantages of static routing**
 - It can backup multiple interfaces/networks on a router
 - Easy to configure for small topologies
 - No extra resources are needed
 - More secure
- **Disadvantages of static routing**
 - Network changes require manual reconfiguration
 - Does not scale well in large topologies

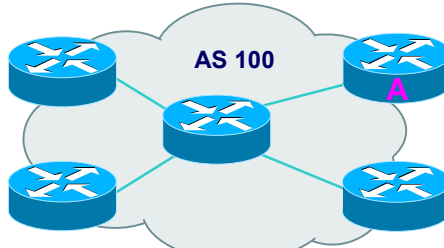
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IGP and EGP

Two groups of Routing Protocols:

- **igp: Interior gateway protocols**
 - Used **within** an autonomous system.
 - Examples: RIP, IGRP, EIGRP, OSPF
- **egp: Exterior gateway protocols**
 - Used to define routes **between** autonomous systems.
 - Examples: EGP, BGP

Autonomous System

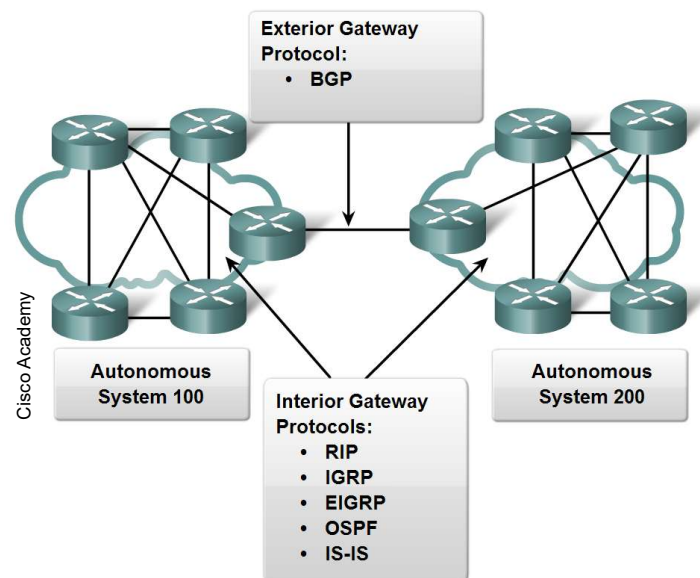
- A set of networks sharing the same routing policy.
 - Identified by ASN - "AS number"
 - Internal connectivity
 - One contiguous unit
- 
- Examples
 - service provider
 - multi-homed customer
 - anyone needing policy discrimination

ASN -Autonomous System Number

- Two ranges
 - 0-65535 (original 16-bit range)
 - 65536-4294967295 (32-bit range - RFC4893)
- Usage:
 - 0 and 65535 (reserved)
 - 1-64495 (public Internet)
 - 64496-64511 (documentation - RFC5398)
 - 64512-65534 (private use only)
 - 23456 (represent 32-bit range in 16-bit world)
 - 65536-65551 (documentation - RFC5398)
 - 65552-4294967295 (public Internet)
- 32-bit range representation specified in RFC5396
 - Defines "asplain" (traditional format) as standard notation

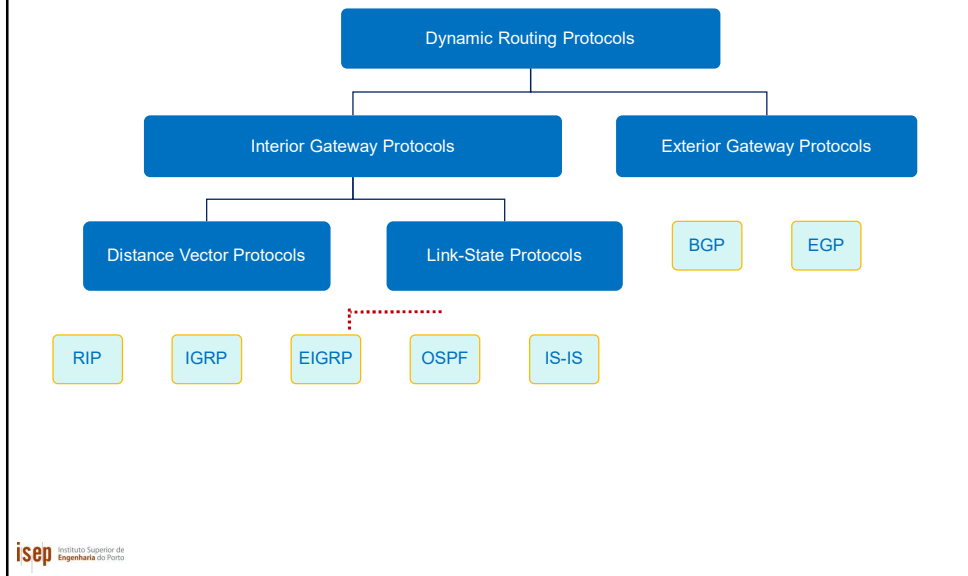
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IGP vs EGP



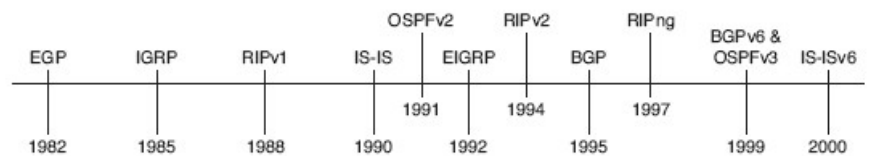
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Dynamic Routing Protocols



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Dynamic Routing Protocols



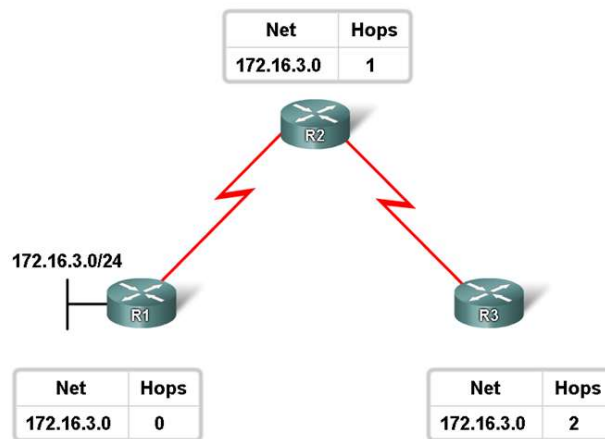
	Interior Gateway Protocols			Exterior Gateway Protocols
	Distance Vector Routing Protocols		Link State Routing Protocols	Path Vector
Classful	RIP	IGRP		EGP
Classless	RIPv2	EIGRP	OSPFv2	BGPv4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	BGPv4 for IPv6

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Routing Protocols Metrics

- Metric: A value used to evaluate routes.
 - Examples: Number of Hops

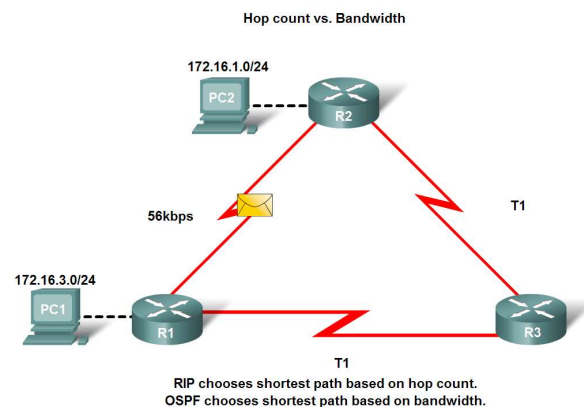


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Routing Protocols Metrics

- Metrics used in IP routing protocols
 - Bandwidth
 - Cost
 - Delay
 - Hop count
 - Load
 - Reliability



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Interior Gateway Protocol (IGP)

Interior Gateway Protocol (IGP)

- Primary goal is optimal connectivity
- Strong distance metrics
- May not have good administrative controls
- Examples
 - RIP, IGRP, HELLO, OSPF
- Two basic methods
 - **Distance Vector** protocols
 - **Link State** Protocols

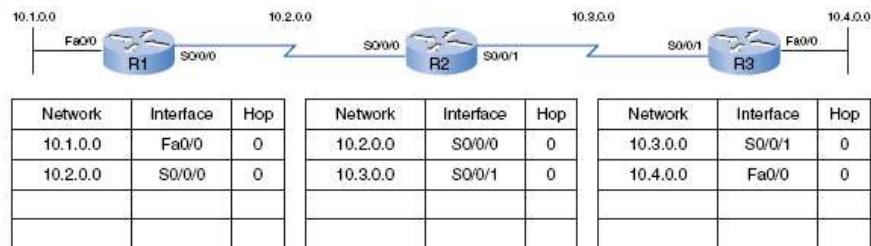
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Distance Vector Protocols

- Usually apply the Bellman-Ford algorithm
- listen to neighboring routers,
- install routes in table,
- use lowest distances,
- advertise **all routes** in table to **neighbors**.
- Very simple.

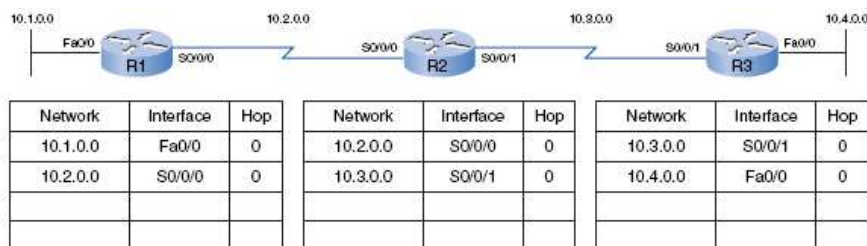
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Cold Start



- Router powers up:
 - Knows nothing about the network topology
 - Does not know devices on the other end of its links.
 - Knows only information saved in NVRAM (startup-config).
- Network discovery
 - part of the process of the routing protocol algorithm
 - enables routers to learn about remote networks.

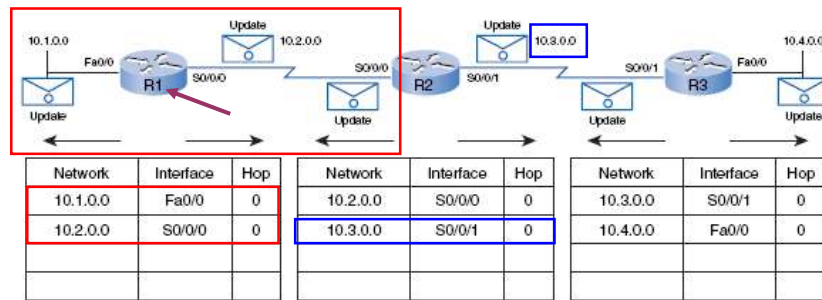
Cold Start



Only knows about it's own networks

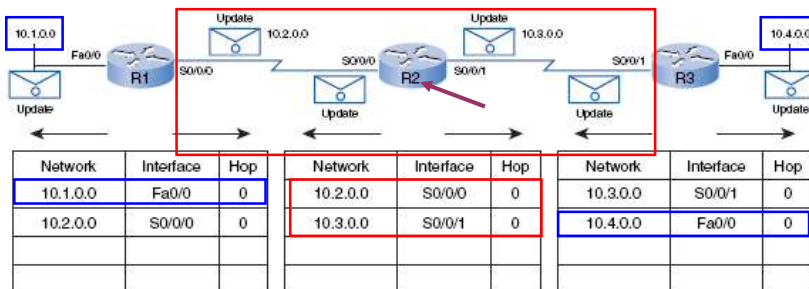
- R1:
 - 10.1.0.0 available through interface FastEthernet 0/0
 - 10.2.0.0 available through interface Serial 0/0/0
- R2:
 - 10.2.0.0 available through interface Serial 0/0/0
 - 10.3.0.0 available through interface Serial 0/0/1
- R3:
 - 10.3.0.0 available through interface Serial 0/0/0
 - 10.4.0.0 available through interface FastEthernet 0/0

Initial Exchange of Routing Information



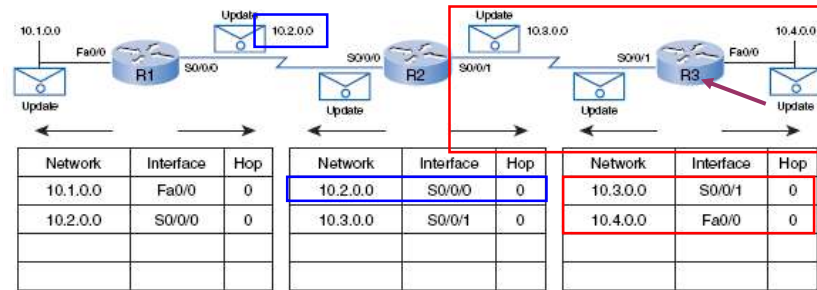
- **R1:**
 - **Sends** an update about network 10.1.0.0 out the Serial 0/0/0 interface with a metric of 1
 - **Sends** an update about network 10.2.0.0 out the FastEthernet 0/0 interface with a metric of 1
 - **Receives** an update from R2 about network 10.3.0.0 on Serial 0/0/0 with a metric of 1
 - **Stores** network 10.3.0.0 in the routing table with a metric of 1

Initial Exchange of Routing Information



- **R2:**
 - **Receives** an update from R1 about network 10.1.0.0 on S0/0/0 with a metric of 1
 - **Stores** network 10.1.0.0 in the routing table with a metric of 1
 - **Sends** an update about network 10.3.0.0 out the S0/0/0 interface with a metric of 1
 - **Sends** an update about network 10.2.0.0 out the S0/0/1 interface with a metric of 1
 - **Receives** an update from R3 about network 10.4.0.0 on S0/0/1 with a metric of 1
 - **Stores** network 10.4.0.0 in the routing table with a metric of 1

Initial Exchange of Routing Information

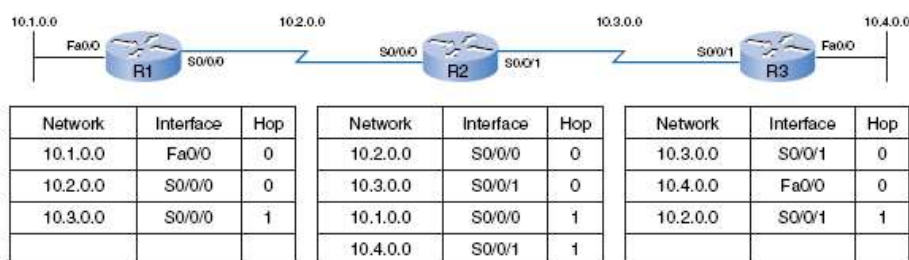


receives

sends

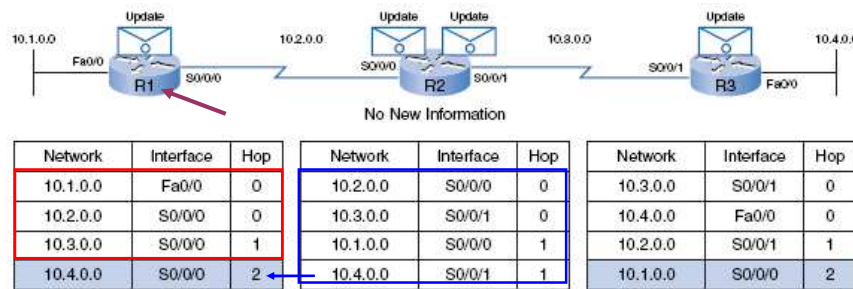
- **R3:**
 - **Sends** an update about network 10.4.0.0 out the S0/0/1 interface with a metric of 1
 - **Sends** an update about network 10.3.0.0 out the Fa0/0 interface with a metric of 1
 - **Receives** an update from R2 about network 10.2.0.0 on S0/0/1 with a metric of 1
 - **Stores** network 10.2.0.0 in the routing table with a metric of 1

Initial Exchange of Routing Information



- **First round of update exchanges:**
 - Each router knows about the networks connected to its direct neighbors.
 - R1 does not yet know about 10.4.0.0
 - R3 does not yet know about 10.1.0.0.
- **Another exchange of routing information is needed.**

Next Exchange of Routing Information



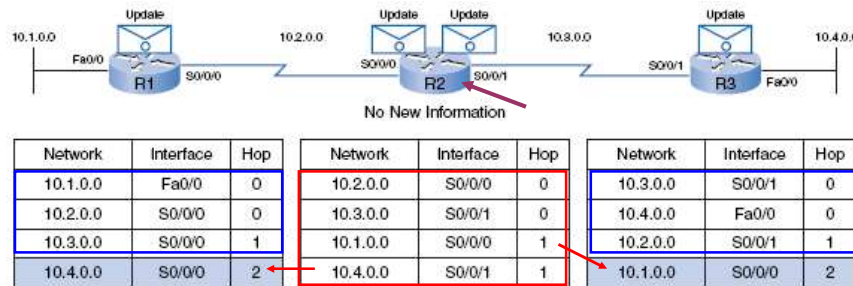
sends

receives

R1:

- **Sends** an update about network 10.1.0.0 out the S0/0/0 interface with a metric of 1.
- **Sends** an update about networks 10.2.0.0 with a metric of 1 and 10.3.0.0 with a metric of 2 out the Fa0/0 interface.
- **Receives** an update from R2 about network **10.4.0.0 on S0/0/0 with a metric of 2.**
- **Stores** network 10.4.0.0 in the routing table with a metric of 2.
- **Same update from R2** contains information about network 10.3.0.0 on S0/0/0 with a metric of 1. There is no change; therefore, the routing information remains the same.

Next Exchange of Routing Information



receives

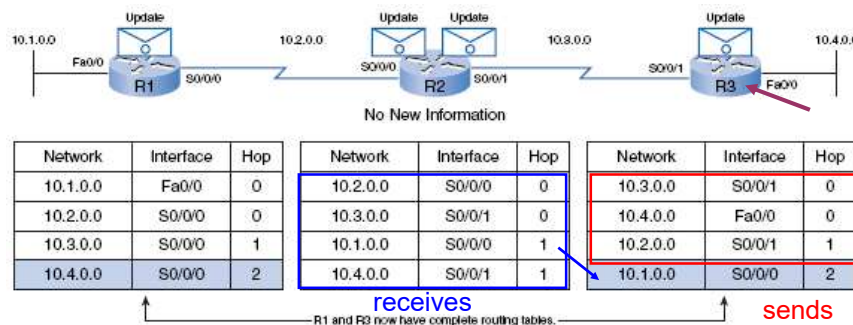
sends

receives

R2:

- **Sends** an update about networks 10.3.0.0 with a metric of 1 and **10.4.0.0 with a metric of 2 out the S0/0/0 interface.**
- **Sends** an update about networks **10.1.0.0 with a metric of 2 (new)** and 10.2.0.0 with a metric of 1 out the S0/0/1 interface.
- **Receives** an update from R1 about network 10.1.0.0 on S0/0/0. There is no change; therefore, the routing information remains the same.
- **Receives** an update from R3 about network 10.4.0.0 on S0/0/1. There is no change; therefore, the routing information remains the same.

Next Exchange of Routing Information



R3:

- **Sends** an update about network 10.4.0.0 out the S0/0/1 interface.
- **Sends** an update about networks 10.2.0.0 with a metric of 2 and 10.3.0.0 with a metric of 1 out the Fa0/0 interface.
- **Receives** an update from R2 about network **10.1.0.0 with a metric of 2**.
- **Stores** network 10.1.0.0 in the routing table with a metric of 2.
- **Same update from R2** contains information about network 10.2.0.0 on S0/0/1 with a metric of 1. There is no change; therefore, the routing information remains the same.

Problems with classic IGP

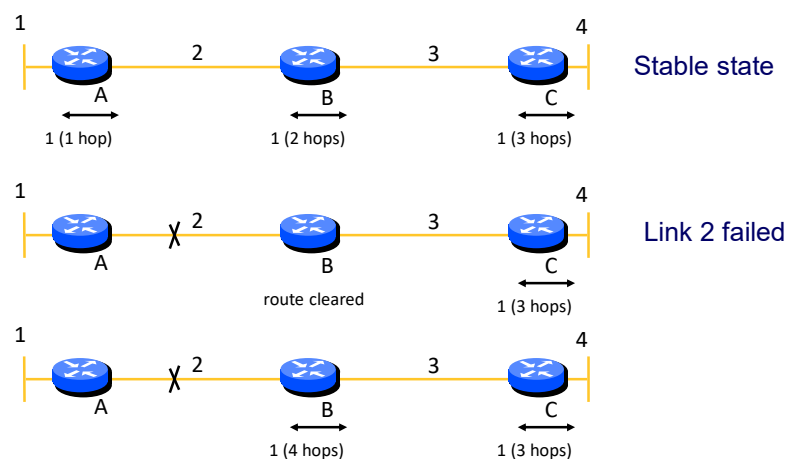
- slow convergence
- count to infinity
- no mask information
 - no CIDR
 - no VLSM
 - no subnet 0

Distance Vector Protocols

- advertisement period
 - entire routing table dumped every n seconds
- timeout period
 - usually 3 times advertisement period
- RIP values are normally 30 and 90 seconds!
- **Slow convergence**

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Count to infinity problem



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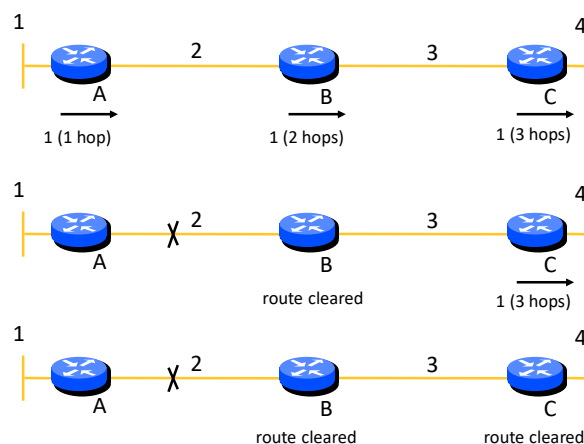
Count to infinity: split-horizon

- Distance Vector Protocols suffer from **Count to Infinity** problem.
- Solution: **Split-Horizon**

- Don't feed selected route back to source
 - no feedback on source interface
 - no feedback to source neighbor

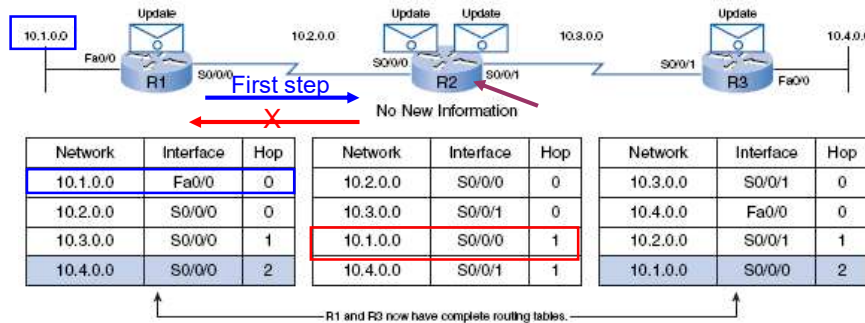
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Count to infinity: split-horizon



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Split Horizon



- Distance vector routing protocols typically apply the **split horizon**.
 - Information is not sent to the interface from which it was received.
- Example
 - R2 would not send an update out S0/0/0 containing the network 10.1.0.0 because R2 learned about that network through S0/0/0.

RIP

RIP: Routing Information Protocol

- Most popular Routing Protocol.
- Interior Gateway Protocol.
- Distance Vector Protocol.
- Only metric is number of hops.
- Maximum number of hops is 15.
- Updates every 30 seconds.
- Doesn't always select fastest path.
- Uses UDP, port 520.

RIP Versions

RIPv1

- No network mask.
- Used broadcasted messages.

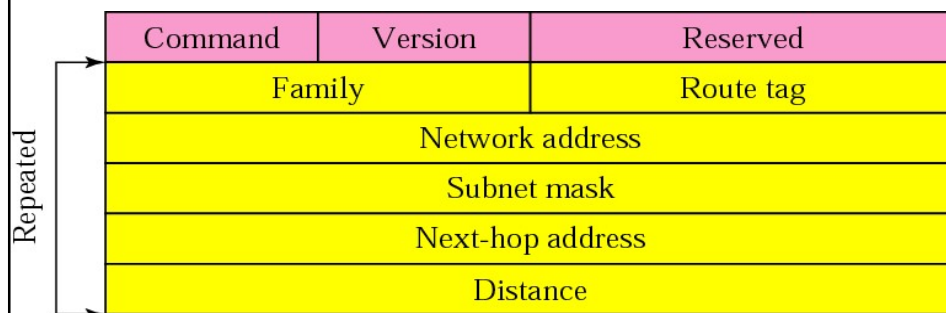
RIPv2

- Classless (CIDR)
- Multicast messages

RIPng

- Supports IPv6
- UDP port 521, multicast group FF02::9.

RIPv2 Packet



IGRP

IGRP: Interior Gateway Routing Protocol

- Cisco proprietary, free use.
- Interior Gateway Protocol.
- Distance Vector Protocol.
- Only IP
- Metric can be composed from:
 - hops,
 - delay,
 - bandwidth,
 - reliability.
 - load,
- Maximum number of hops is 255.
- Updates every 90 seconds

Link-State IGP

- Interior Gateway Protocols
- Using Link-State Information sharing
- Examples:
 - OSPF: Open Shortest Path First
 - IS-IS: Intermediate System to Intermediate System

Link state protocols

- Information about **adjacency** sent to **all** routers.
- each router builds a topology database.
- A "shortest path" algorithm (*Dijkstra*) is used to find best routes.
- Converge as quickly as databases can be updated.